

Hypothesis testing example

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```
library(report)
set.seed(20220622)
knitr::opts_chunk$set(echo = TRUE)
```

Hypothesis testing example

This example demonstrates two-group hypothesis tests using base R functions like `t.test` and `wilcox.test`. It also uses the `report` package for easy reporting of results.

Enter data

We will simulate random data for two experimental conditions, A and B.

```
dt_A <- rnorm(n = 10, mean = 1.0, sd = 0.3)
fct_A <- rep(x = 'A', times = length(dt_A))

dt_B <- rnorm(n = 10, mean = 1.5, sd = 0.3)
fct_B <- rep(x = 'B', times = length(dt_B))

dt <- data.frame(
  value = c(dt_A, dt_B),
  condition = c(fct_A, fct_B)
)

summary(dt)
```

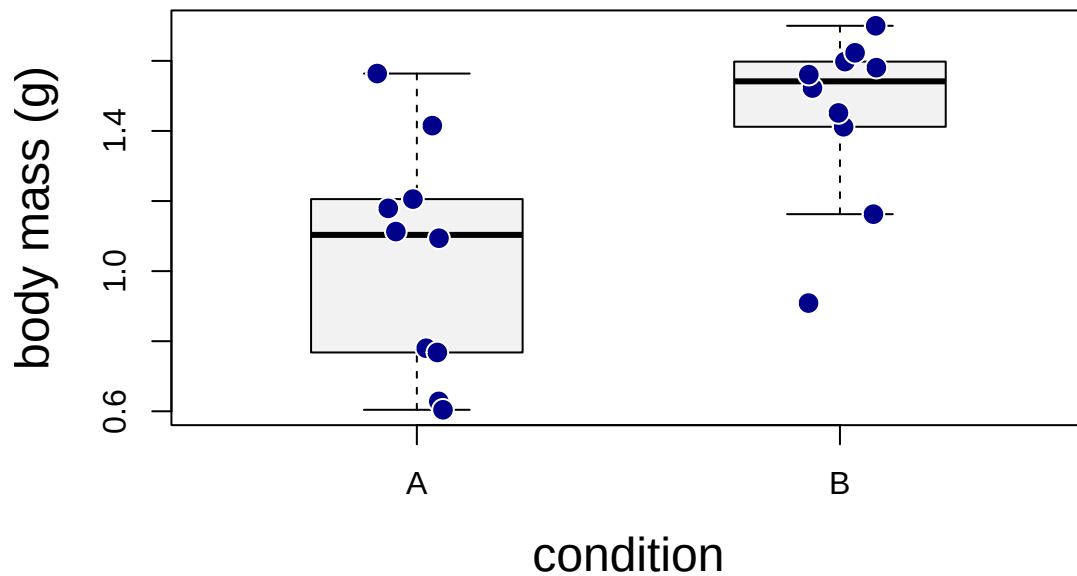
```
##      value      condition
## Min.   :0.6043   Length:20
## 1st Qu.:1.0476   Class :character
## Median :1.3087   Mode  :character
## Mean   :1.2435
## 3rd Qu.:1.5615
## Max.   :1.7001
```

Inspect the data

We will visualise the distribution with a boxplot and raw data points.

```
boxplot(value ~ condition,
        data = dt,
        ylab = 'body mass (g)',
        xlab = 'condition',
        ylim = range(dt$value),
        pars = list(boxwex = 0.5),
        outline = FALSE,
        col = 'gray95',
        cex.lab = 1.5)

stripchart(value ~ condition,
           data = dt,
           vertical = TRUE,
           add = TRUE,
           method = 'jitter',
           pch = 21,
           bg = 'darkblue',
           col = 'white',
           cex = 1.5)
```



Perform a Student's t-test

We ask whether `condition` predicts `value`, using a two-sided Student's t-test with equal variances.

```
tt <- t.test(
  formula = value ~ condition,
  data = dt,
  alternative = "two.sided",
  var.equal = TRUE
)

print(tt)
```

```
##
## Two Sample t-test
##
## data: value by condition
## t = -3.2324, df = 18, p-value = 0.004621
## alternative hypothesis: true difference in means between group A and group B is not equal to 0
## 95 percent confidence interval:
## -0.6878178 -0.1459245
## sample estimates:
## mean in group A mean in group B
## 1.035062 1.451933
```

```
suppressWarnings(report(tt))
```

```
## In the report() function, for htest objects, you can try providing the data argument
## manually, e.g., report(x, data = data).

## Effect sizes were labelled following Cohen's (1988) recommendations.
##
## The Two Sample t-test testing the difference of value by condition (mean in group A = 1.04, mean in
## group B = 1.45) suggests that the effect is negative, statistically significant, and large
## (difference = -0.42, 95% CI [-0.69, -0.15], t(18) = -3.23, p = 0.005; Cohen's d = -1.52, 95% CI
## [-2.56, -0.46])
```

Welch's t-test for unequal variances

Now we simulate a dataset where the two groups have different variances and apply Welch's t-test.

```
dt_unequal <- within(dt, {
  value[condition %in% 'B'] <- rnorm(n = 10, mean = 1.5, sd = 0.15)
})
```

```
summary(dt_unequal)
```

```
##      value      condition
## Min.   :0.6043   Length:20
## 1st Qu.:1.1081   Class :character
## Median :1.4116   Mode  :character
## Mean    :1.2885
## 3rd Qu.:1.5827
## Max.    :1.6866
```

```
tt_welch <- t.test(
  formula = value ~ condition,
  data = dt_unequal,
  alternative = "two.sided",
  var.equal = FALSE
)
```

```
print(tt_welch)
```

```
##
## Welch Two Sample t-test
##
## data: value by condition
## t = -4.5225, df = 11.884, p-value = 0.0007159
## alternative hypothesis: true difference in means between group A and group B is not equal to 0
## 95 percent confidence interval:
## -0.7514202 -0.2624398
## sample estimates:
## mean in group A mean in group B
##      1.035062      1.541992
```

```
suppressWarnings(report(tt_welch))
```

```
## In the report() function, for htest objects, you can try providing the data argument
## manually, e.g., report(x, data = data).
```

```
## Effect sizes were labelled following Cohen's (1988) recommendations.
```

```
##
```

```
## The Welch Two Sample t-test testing the difference of value by condition (mean in group A = 1.04,
## mean in group B = 1.54) suggests that the effect is negative, statistically significant, and large
## (difference = -0.51, 95% CI [-0.75, -0.26], t(11.88) = -4.52, p < .001; Cohen's d = -2.62, 95% CI
## [-4.14, -1.05])
```

Non-normal distributions and the Mann-Whitney test

For data that may violate the normality assumption, we simulate an asymmetric distribution and compare groups using Welch's t-test and the Mann-Whitney rank sum test.

```
dt_un_A <- rnorm(n = 10, mean = 1.0, sd = 0.3) + rpois(n = 10, lambda = 3.4)
fct_un_A <- rep(x = 'A', times = length(dt_un_A))

dt_un_B <- rnorm(n = 10, mean = 3.5, sd = 0.6) + rpois(n = 10, lambda = 3.4)
fct_un_B <- rep(x = 'B', times = length(dt_un_B))

dt_unknown <- data.frame(
  value = c(dt_un_A, dt_un_B),
  condition = c(fct_un_A, fct_un_B)
)

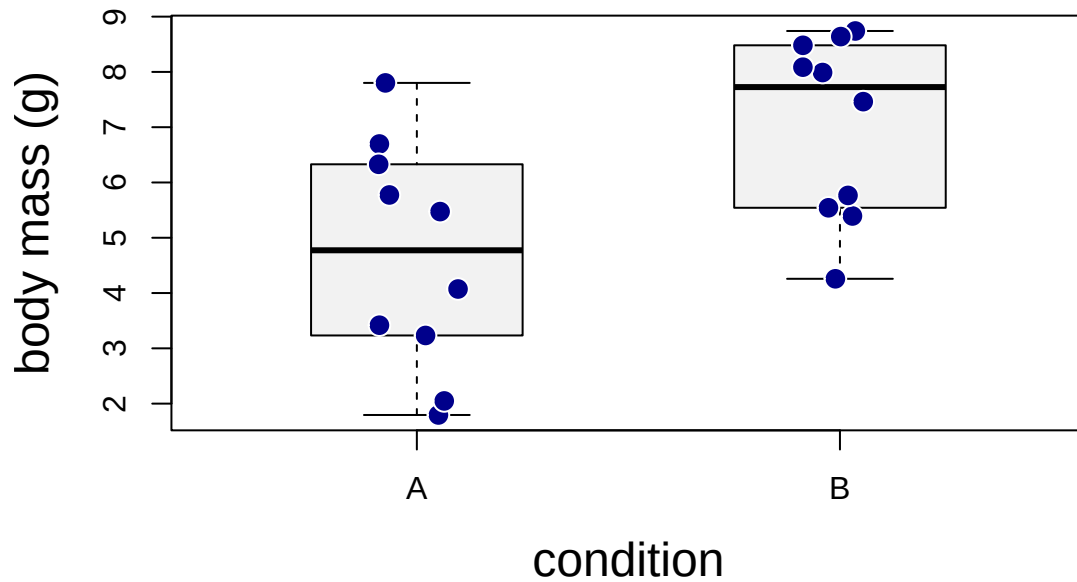
summary(dt_unknown)
```

```
##      value      condition
## Min.   :1.794   Length:20
## 1st Qu.:4.212   Class :character
## Median :5.770   Mode  :character
## Mean    :5.850
## 3rd Qu.:7.848
## Max.    :8.740
```

```
boxplot(value ~ condition,
  data = dt_unknown,
  ylab = 'body mass (g)',
  xlab = 'condition',
  ylim = range(dt_unknown$value),
  pars = list(boxwex = 0.5),
  outline = FALSE,
  col = 'gray95',
  cex.lab = 1.5)

stripchart(value ~ condition,
  data = dt_unknown,
  vertical = TRUE,
  add = TRUE,
  method = 'jitter',
```

```
pch = 21,
bg = 'darkblue',
col = 'white',
cex = 1.5)
```



```
mean_unknown <- aggregate(
  value ~ condition,
  data = dt_unknown,
  FUN = function(x) {
    sum_stat <- mean(x) + 1.96 * c(0, -1, 1) * sd(x) / sqrt(length(x))
    names(sum_stat) <- c('mean', 'lower CI', 'upper CI')
    return(sum_stat)
  }
)

print(mean_unknown)
```

```
##   condition value.mean value.lower CI value.upper CI
## 1         A   4.664204      3.396375      5.932033
## 2         B   7.035760      6.023326      8.048195
```

```
tt_unknown <- t.test(
  formula = value ~ condition,
  data = dt_unknown,
  alternative = "two.sided",
  var.equal = FALSE
)
```

```

print(tt_unknown)

##
##  Welch Two Sample t-test
##
## data:  value by condition
## t = -2.8649, df = 17.16, p-value = 0.01066
## alternative hypothesis: true difference in means between group A and group B is not equal to 0
## 95 percent confidence interval:
##  -4.1168030 -0.6263092
## sample estimates:
## mean in group A mean in group B
##           4.664204           7.035760

mww <- wilcox.test(
  value ~ condition,
  data = dt_unknown,
  alternative = "two.sided"
)

print(mww)

##
##  Wilcoxon rank sum exact test
##
## data:  value by condition
## W = 19, p-value = 0.01854
## alternative hypothesis: true location shift is not equal to 0

```

Rank-transformed visualisation

The plot below shows both the original values and the rank-transformed values for the two conditions.

```

par(mfrow = c(1, 2))
with(dt_unknown, {
  hist(value[condition %in% 'A'],
        breaks = 1e2,
        col = 'blue',
        xlim = range(value),
        xlab = 'value',
        main = 'original values')
  hist(value[condition %in% 'B'],
        breaks = 1e2,
        col = 'red',
        add = TRUE)
  hist(rank(value)[condition %in% 'A'],
        breaks = 1e2,
        col = 'blue',
        xlim = c(0, max(rank(value))),
        xlab = 'value',
        main = 'rank-transformed')
}

```

```
hist(rank(value)[condition %in% 'B'],
     breaks = 1e2,
     col = 'red',
     add = TRUE)
})
```

